

DIPIKA KHULLAR

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EDUCATION

University of California, Berkeley	B.A Computer Science, Major B.A. Data Science, Minor	December 2020
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Awards: Berkeley Leadership Award

PROFESSIONAL EXPERIENCE

MATS Scholar

Berkeley, CA

Google Deepmind Stream

Jan 2026 — present

- Developing a model organism for obfuscated reasoning that studies how broad reasoning training (across math, coding, and agentic tasks) can reduce monitorability on out-of-distribution safety-critical tasks.

Anthropic

Berkeley, CA

Research Fellow

May 2025 — present

- Researched deception and self-sycophancy in LLMs, developing methodologies to measure inflated self-ratings in harmfulness judgments and benchmarking failure modes across models.
- Steering models to be more efficient and faithful in their reasoning traces, reducing unproductive backtracking while preserving accuracy.

SPAR Research Fellow

Berkeley, CA

Research Fellow

Jan. 2025 — May 2025

- Published steering-vector interventions that edit LLM activations to reduce inefficient backtracking, achieving more accurate and cost-efficient reasoning traces.

Amazon AGI

Santa Clara, CA

Applied Scientist II

Mar. 2022 — May 2025

- Led end-to-end pretraining data curation for large-scale multimodal models, spanning petabytes of web-crawled text and vision data (HTML, native-text and OCR-extracted PDFs). Analyzed error taxonomies (syntax, reasoning, planning failures) to guide data curation and sampling strategies, and led structured content extraction to surface high-quality code, math, and tabular data from noisy web and document corpora. Designed internal evaluations for reasoning tasks to diagnose generalization vs. memorization failures in pretrained models.
- Identified, evaluated, and benchmarked high-quality data sources for Bedrock foundation models through systematic data value assessments and ablation studies; released PILEv2 and automated future PILE data source refreshes.
- Led the proposal, OCR research, modeling, and engineering for a PDF structured data extraction pipeline applied to AmazonBot-fetched web documents, enabling large-scale extraction of structured signals from scanned and semi-structured PDFs for Nova model pretraining.
- Productionized data pipelines for filtering, deduplication, quality scoring, and mixture weighting, improving downstream performance on code, math, and reasoning benchmarks by 12%.
- Developed synthetic data generation and labeling pipelines for robotics foundation models (AWS GenAI Innovation Center), supporting perception and decision-making tasks for early-stage robotics startups; validated effectiveness via ablations and downstream task performance.
- Proposed and developed a diffusion-based synthetic image generation method for inserting rare objects into real backgrounds using text-conditioned diffusion and mask-based blending, improving object detection training on scarce real-world road scene categories.

Qualcomm Corporate Research and Development

San Diego, CA

Research Engineer

Dec. 2020 — Mar. 2022

- Worked with the framework integration team to open source examples and benchmarks for deep learning model optimization toolkits and federated learning library
- Developed 8 bit quantization and compression methods for CNNs, transformers, and other state of the art models. Experiment with heuristics to identify output channels that will cause a high error term post quantization. Work open

sourced as part of [AIMET](#).

- Created a visualization tool, exposing statistics and data used for making model quantization and compression decisions. Analyzed compression performance, visualized evaluation scores and compression ratios to understand the candidates selected by greedy selection algorithms

Square

Machine Learning Engineering Intern

San Francisco, CA

Sept. 2020 — Dec. 2020

- Utilized vector embeddings for categorical data to increase the predictive power of XGBoost loan model for Square Capital. Explored unsupervised learning techniques for misclassified business types
- Integrated new computational flow and modeling framework with the Square Capital loan platform software

Apple CoreML

Machine Learning Engineering Intern

Cupertino, CA

May 2020 — Aug. 2020

- Prototyped and experimented with neural architecture search (NAS) methods for CreateML sound classification
- Exposed a parameterizable MLP head for the [image classifier in CreateML](#), providing a higher capacity alternative to logistic regression. Determined optimal parameters for this new image classification model
- Profiled [sound](#) and [image](#) classifier performance improvements using a multilayer perceptron (MLP) classifier head over logistic regression with larger dataset sizes
- Determined an automatic selection of classifier heads for both sound and image classification tasks based upon the number of training examples, classes, and balance of a dataset

RESEARCH PUBLICATIONS

[Making VLMs More Robot-Friendly: Self-Critical Distillation of Low-Level Procedural Reasoning](#): Poster, EMNLP 2025

[Self-Attribution Bias in Risk and Correctness Judgments](#): Neurips 2025 Multiturn interactions workshop, in submission ICML 2026

[Making Language Models Version Aware](#): (In submission, ARR)

[Reasoning Models Reason Inefficiently](#): *Neurips 2025: Workshop on Efficient Reasoning*

[Kaleidoscope: In-language Exams for Massively Multilingual Vision Evaluation](#): in submission ICML 2026

[Synthetic Data Generation for Scarce Road Scene Detection Scenarios](#): *Poster, Workshop on Synthetic Data Generation with Generative AI, Neurips 2023*

[Improved Few-Shot Image Classification Through Multiple-Choice Questions](#): *WACV 2024, Poster, Amazon Computer Vision Conference 2023*

P84597: Fully automated and controlled Virtual Product Placement in Images and Videos : US Patent

ACADEMIC EXPERIENCE

Community Researcher: Eleuther AI

Jan 2025 — present

- *Estimating volume basins in parameter space, quantifying how much perturbation weights could tolerate before significantly degrading performance. Advised by Nora Belrose*

Undergraduate Researcher: Assistant AI Professor Canny's Group

August 2020 — March 2022

- Working in Professor John Canny's lab on new techniques for OCR, learning methods in order to support a video-to-text translation pipeline
- Investigate and design representation learning algorithms that extract general and meaningful latent features and profile them against current video captioning systems

Research Assistant: Purdue University

August 2020 — May 2021

- Understanding comparison networks and their structure, with the goal of improving their performance.

BLOG POSTS

[Create Amazon SageMaker models using the PyTorch Model Zoo](#)

[AWS Machine Learning Blog](#)

Dipika Khullar, Marcelo Aberle, Ninad Kulkarni, and Yash Shah

[Build Streamlit apps in Amazon SageMaker Studio](#)

[AWS Machine Learning Blog](#)

Dipika Khullar, Marcelo Aberle, and Yash Shah

INVITED TALKS

Domain-Compatible Synthetic Data Generation for Infrequent Object Detection

Amazon Science

SERVICE

Amazon Computer Vision Conference 2023 (ACVC)

Peer Reviewer

Winter Conference on Applications of Computer Vision 2023 (WACV)

Peer Reviewer

Amazon Research Award*

Grant Reviewer Committee

* Reviewed research proposals sent in by PhD students and faculty to determine funding for the [Amazon Research Awards](#)